

UBT202: MICROBIOLOGY

L	T	P	Cr
3	0	0	3.0

Course Objective: To provide fundamental understanding of the microbial world, basic structure and functions of microbes, metabolism, nutrition, their diversity, physiology and relationship to environment and human health. To impart practical skills of isolation and manipulating conditions for their propagation.

Syllabus

History and classification: Brief history on development and scope of microbiology, characterization, classification and identification of microorganisms, numerical taxonomy and molecular approaches, microscopic examination of microorganisms, bacterial staining, simple and differential staining

Morphology and fine structure of microorganisms: Prokaryotes and eukaryotes, bacterial diversity, bacterial cell structures, Gram positive and Gram-negative bacteria, morphological features, cell structure and major characteristics of cellular (bacteria, fungi, algae, protozoa) and acellular (viruses) organisms

Cultivation and cultural characterization of microorganisms: Nutritional and physical requirements of autotrophs, heterotrophs, chemotrophs and lithotrophs, types of culture media, enumeration of microbial populations, pure culture and cultural characteristics
Microbial Growth: Modes of cell division, normal growth cycle, and quantitative measurement of growth, growth curve, synchronous growth and continuous culture, factors affecting growth, sporulation, Maintenance and preservation of microbial cultures and its importance, culture banks

Microbial Physiology: Bacterial metabolism, energy production, respiration, intermediate metabolism, fermentation and photosynthesis

Microorganisms and Diseases: Major diseases caused by different microorganism in human, animals and plants

Microbial Control: Physical and chemical agents for control of microbial growth, their mode of action, sterilization, disinfectants and antiseptics, chemotherapeutic agents, antibiotic susceptibility test

Course Learning Objectives (CLO)

The students will be able to:

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

1. Define the science of microbiology, its development and importance in human welfare.
2. Describe historical concept of spontaneous generation and the experiments performed to disprove.
3. Describe some of the general methods used in the study of microorganisms.
4. Recognize and compare structure and function of microbes and factors affecting microbial growth.
5. Demonstrate aseptic microbiological techniques in the laboratory and check sources of microbial contamination and their control.

Text Books

1. Pelczar Jr., M.J., Chan, E.C.S. and Krieg, Noel R., Microbiology, Shree Hari Publications (2021).
2. Stanier, R.Y., Ingraham, J.L. and Wheelis, M.L., General Microbiology, MacMillan (2007) 5th ed.
3. Madigan, M. T. and Martinko, J. M., Brock Biology of Microorganisms, Pearson Publication (2017) 14th ed.

Reference Books

1. Tortora, G.J., Funke, B.R., and Case, C.L., Microbiology: An Introduction, Pearson Education (2019) 4th ed.
2. Willey, J. and Sherwood, L., Prescott's Microbiology, Asia Higher Education Science Microbiology (2017)

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30

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